Arnav Tiwari

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Professional Summary

Full-Stack Software Engineer specializing in architecting and deploying scalable, AI-powered web applications using React, Next.js, Node.js, and PostgreSQL. Proven track record in designing high-performance **RESTful APIs**, optimizing **relational and NoSQL data models** for high-throughput environments, and implementing **multi-agent LLM orchestration** using **LLM Integration (Gemini, OpenAI, Claude)**. Expert in containerized deployments via **Docker** and automating **CI/CD pipelines** to ensure system reliability and production-ready delivery at scale.

Education

Indian Institute of Information Technology Nagpur (IIIT Nagpur) *November 2022 – July 2026*
BTech in Computer Science Engineering

Technical Skills

Programming Languages: C, C++, Java, Python, JavaScript, TypeScript
Frontend Development: React.js , Next.js, Redux, Tailwind CSS, Framer Motion, Bootstrap
Backend & APIs: Node.js, Express.js, Django, RESTful APIs, Authentication & Authorization (JWT, OAuth 2.0, bcrypt), Razorpay API
Databases & Data Modelling: PostgreSQL, MySQL, MongoDB, Drizzle ORM, Mongoose, Vector DBs
Gen AI & Automation: LLM App Dev, RAG, Google Gemini API, Prompt Engineering, n8n Workflow Automation
Cloud, DevOps & Tooling: Docker, Kubernetes, CI/CD, GitHub Actions, Vercel, Netlify, Git, GitHub, Postman

Professional Experience

Software Developer Intern *July 2025 – Present*
Hitachi India Pvt Ltd. | Nagpur, Maharashtra

- Developed and optimized Single Line Diagram (SLD) modules for power system visualization, improving analysis efficiency and system clarity.
- Worked with IntervalZero RTX64 RTOS to run Hitachi applications without depending on Windows OS.
- Developed Frontend in **React** and backend in **Django** of HICADS Projects - DPS (Digital Permit System), RCTMS (Roll Coolant Tank Monitoring System), PTMS (Pickling Tank Monitoring System), TTMS (Truck Turnaround Monitoring System), and ICQIS (Intelligent Coating Quality Improvement System) as solutions for Steel Processing Industries.
- Automated the Customer Requirement Specification (CRS) analysis workflow for Hitachi using n8n, achieving a **93% reduction in manual processing time** and project document generation (BNB, Proposal, Risk Assessment).

Full Stack Developer Intern *May 2025 – July 2025*
Payatu Security Consulting Pvt. Ltd. | Pune, Maharashtra

- Built responsive **Next.js** + **SCSS** interfaces, reducing page load time by 25% and improving UI maintainability.
- Developed and optimized MySQL-backed **Nodejs REST APIs**, improving backend response time by 40%.
- Developed a Real time logger application for displaying and storing logs of a RBAC wifi admin dashboard web application.

Projects

Autopilot- AI Powered Project Planning Platform
Next.js, TypeScript, TailwindCSS, Express.js, PostgreSQL, Google Gemini, Groq SDK, OAuth 2.0

- Architected a decoupled, multi-agent AI system with 4 specialized agents (Google Gemini, Groq SDK) to automate project planning from natural language input, generating requirements, tech stack recommendations, risk assessments, and task breakdowns.
- Designed and implemented a RESTful API layer to orchestrate agent coordination, process project specifications, and return structured execution plans with confidence scores and cost estimations.
- Delivered production-ready features including OAuth 2.0 authentication, PostgreSQL database integration, multi-format document export (PDF, DOCX, JSON), internationalization (i18n), and a WCAG-compliant dark-mode UI.

AI-Powered Content Generator Platform
Next.js, Clerk, Drizzle ORM, PostgreSQL, Tailwind CSS,TypeScript, Razorpay, Google Gemini API

- Built a scalable LLM-powered SaaS platform integrating Google Gemini API with optimized prompt engineering to generate high-quality content across 17+ dynamic templates, reducing content creation time by 70% compared to manual workflows.
- Engineered a PostgreSQL backend with Drizzle ORM to support high-throughput workloads, sustaining 10,000+ monthly AI generations with schema-level data integrity and query optimization.
- Implemented end-to-end user authentication, role-based access control, and subscription payment processing using Clerk and Razorpay, resulting in a 35% improvement in paid conversion rate.

Book Store – Personalised Book Collection Platform
React, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, JWT

- Built a full-stack web application with a component-driven React frontend (15+ reusable components) and a Node.js/Express.js REST API backend, implementing CRUD operations, stateless JWT authentication, and bcrypt-based password hashing.
- Engineered MongoDB data models with schema validation and file upload handling via Multer middleware, enforcing data integrity constraints and enabling reliable binary asset storage.

Achievement

- Achieved **97.6 percentile (AIR 22,007)** in JEE Mains 2022 among 1.1M+ candidates
- **1st Runner Up – ArtPark Codeforge Hackathon** (Unstop) — competed against teams across India in a software development challenge.
- Solved **350+ problems** on **LeetCode** across arrays, dynamic programming, and graph algorithms.